

Magnetic Matter Movers Space Transportation Architecture

Minimum-Chemical Launch Concept Study

Compiled July 15, 2026 — physics checks, subsystem sizing, cost models, and the decision trail Internal program designation: MTS — Magnetic Transportation Systems**

Executive Summary

A three-system launch portfolio plus one orbital node, sharing magnetics-and-power DNA, with chemical rockets eliminated except as a crew abort motor:

1. **Coilgun freight line** — 4.9 km track at 1,000 g, bulk mass to LEO at ~\$150/kg (direct-to-escape variant: 8.7 km, ~\$190/kg)
2. **Maglev + laser-thermal cargo** — 6.8 km track (30 g) + 475 MW beam, delicate cargo to LEO, \$323–3,800/kg depending on flight rate
3. **Maglev + laser-thermal crew** — 38 km track (3 g) + 3.8 GW beam, 6-seat vehicles, \$0.6–2.4M/seat at maturity
4. **LEO dock + tug fleet** — slugs become ships; tugs (beamed-perigee-kick or electric) add the final 3.2 km/s to escape

Total capital ladder: \$0.1B pilot → \$4–12B cargo → \$23–84B crew. Every phase generates revenue and de-risks the next. Marginal costs are electricity: \$11/kg (beamed) and ~\$3.6/kg (gun) in energy alone.

Epistemic Convention (owner standard, adopted Jul 20, 2026)

Claims in this study carry three tiers: **[E] established** (textbook physics, published data), **[I] inference** (first-cut analysis, scaling laws, our sims — method-limited), **[S] speculation** (demand models, far-horizon concepts). Blanket assignments unless marked otherwise: Physics Ground Rules = [E]. Sim results = [I] (v0/v1 are first-cut: limited Monte Carlo, reduced-order blooming). **All dollar figures = [I]-grade approximations from memory-sourced cost curves, NOT verified quotes — and they sit atop 5+ step inference chains (payload fraction × capex × rate × amortization) where errors compound; treat every \$/kg as ±2× until work item 22 (citation + uncertainty pass) closes.** Demand/market claims = [S]. *Amended Jul 20, 2026: the Cost Calibration Doctrine (extension, end of document) governs how dollar figures are stated and judged — bands with honest midpoints plus a 3× survivability test; the item-22 citation pass first cut (this directory) sources the headline rows, and its corrections are applied in this commit.* Governing equations for checkable math: rocket equation $\Delta v = v_e \ln(m_0/m_1)$ · jet power $P = T v_e / 2$ · magnetic pressure $B^2 / 2\mu_0$ · Inductrack $L/D \approx v/w$ · Bradley-Herrmann N_D scaling · $F = BIL$ — implementations in `mmm_waveoptics_v0/v1.py` and the conversation-run models.

Physics Ground Rules

Space is fast, not high. Three destinations, three price tags: touch 100 km and fall back — 1.4 km/s; stay in LEO — 7.8 km/s; leave Earth (escape) — 11.2 km/s ($= \sqrt{2} \times$ orbital). Altitude to 100 km is 1 MJ/kg; orbital kinetic energy is 30 MJ/kg. Any altitude-first scheme (balloons, towers) attacks 3% of the problem.

The atmosphere is a 17% tax. Gravity + drag + steering losses ≈ 1.6 of the 9.4 km/s LEO budget. Worth harvesting (evacuated tube, mountain exit, thrust above the dense air) — never worth building an architecture around.

The rocket equation lives at the top of the velocity budget. Shaving the first 2 km/s saves ~8% of propellant; raising exhaust velocity attacks the exponent. Ground-assist helps; changing the engine (I_{sp} 350 → 900) transforms (payload fraction 2% → 21%).

Every technology is regime-locked. Electric thrusters, power beaming, and Earth's magnetic field are all useless during ascent and excellent after orbit. Chemical wins ascent by *power density*, not efficiency — the propellant is the power plant. To launch without chemistry, either leave the power plant on the ground (beam) or deliver all velocity on the ground (track/gun).

G-force sorts the passengers permanently. Humans (≤ 3 g) can never take their velocity from the ground: direct-to-escape would need a 2,655 km track. Crew requires sustained thrust over minutes. Hardened cargo at 1,000 g needs 4.9–8.7 km. This is a law, not an engineering gap.

Ballistic coefficient is destiny in the atmosphere. A dense, slender 1,000 kg slug from a 4 km exit loses 516 m/s and ~1 kg of ablator crossing the whole atmosphere at 13 km/s. A flat plate (the Pascal-B manhole cover) becomes a reverse meteor.

Design Filters (Why Branches Were Cut)

1. **No step-function infrastructure.** Anything that must be 100% complete before delivering 1% of value will consume the program. Prefer systems that scale continuously and earn on every phase.
2. **Put complexity where the wrenches are.** Vehicle dumb and cheap; intelligence, power, and maintenance on the ground.
3. **Space assets must be thin, passive, and optional.** A dead relay mirror is a degraded day; a cut tether is a dead program.

4. **Non-contact magnetic force wherever surfaces meet at speed.** Inductrack levitation, coilgun (not railgun) launch, magnetic tug docking aids.

Rejected Branches (and the numbers that killed them)

Concept	Fatal number	Verdict
Momentum-exchange tether	15–25 t infrastructure per t payload; catch = hypersonic rendezvous, meter tolerance, seconds; single structure in the debris belt	Step-function megastructure; bootstrap paradox
Fully electric launch	T/W > 1 needs ~100 MW-class jet power per tonne; power plant 10–500× vehicle mass; scale-invariant	Specific-power wall (~10–100× beyond batteries/reactors)
EAD as launch stage	Thrust density 2–20 N/m ² → 2,551 m ² of electrode for a 1 t vehicle; no air above ~50 km; thrust decays with airspeed	Regime-locked to atmospheric flight (per Gomez Vega framing)
Harnessing Earth's magnetic field	50 μT field = 1 mPa magnetic pressure; dipole levitation of 1 t needs 4×10 ¹⁴ A·m ² ; static field carries no energy	Cannot amplify a field you didn't generate; no reaction partner. Exception: electrodynamic tethers in orbit (3 N from 10 km/10 A → 1,555 m/s per month, propellant-free)
100 km "zipper tower"	Altitude = 3% of the job; NdFeB self-support height 1.1 km; erection mass 50,000+ t; magnetic joints have no bending stiffness	Compression + mass + hinging. Salvage: the instinct (magnetic structure held by fields) is the launch loop / space fountain family — dynamic support via momentum flux — itself rejected on filter #1
Space-based power generation	950 MW platform = 3,200 t solar + 2,400 t radiators (waste heat has no pond in space); burst variant still ~1,500 t (ISS = 420 t)	Bootstrap paradox, heavier than the tether
Ground-only slant beaming (full burn)	Isoplanatic patch ~16 μrad vs vehicle crossing 8,571 μrad/s; thermal blooming N _D = 270 at end-of-burn slant	Adaptive optics structurally cannot track the target; works only for first ~⅓ of burn
Microwave (28 GHz) instead of laser	Published system estimate: ~80 MW beam + 90–175 m antenna for an 8 kg payload ≈ 10,000 MW/t — ~100× our laser figure (95 MW/t) — because 10,000× longer wavelength wrecks diffraction; pulsed air-plasma thrusters also self-shield (residual plasma absorbs the next pulse)	Cheap watts (fusion-industry gyrotrons), impossible optics; CW solid-HX laser avoids plasma timing entirely. Their tracking data imported below

Selected Architecture

Systems are lettered in build order — A is built first, and each system's flight heritage is the entry condition for the next.

System A — Coilgun Freight (build first, with the pilot)

- **Staged-to-LEO (primary):** 9.8 km/s muzzle, 4.9 km track, 1,000 g, 48 GJ/shot (36 GW pulse). Slug: 180 kg shell + 190 kg circularization kick → **~630 kg to LEO, ~\$150/kg**
- **Direct-to-escape (variant):** 13.05 km/s muzzle, 8.7 km track, 85 GJ (64 GW pulse); 670 kg cargo to escape at **\$190/kg** (5,000 shots/yr) — vs published chemical pricing: ~\$3–7k/kg to LEO (2026 Falcon-class list/rideshare), \$5–10k/kg escape-class [sourced, item-22 pass]
- **Atmosphere transit (4 km mountain exit):** 516 m/s toll, peak 109 kW/cm² for <1 s, ~1 kg ablated (carbon-phenolic, 3× margin carried)
- **Gating tech: pulsed power.** 36–64 GW for ~1.3 s = flywheel motor-generator / compensated pulsed-alternator farms (batteries and capacitors both disqualify). ~1,500–2,700× (variant-dependent) the largest EM launcher built (Navy/GA ~33 MJ class) — the biggest extrapolation in the portfolio
- **Coilgun, not railgun:** no contact at 10–13 km/s. 25 MW feed → 20+ shots/day. Cargo: water, cryogenics, metals, propellant, artillery-grade electronics (15,000 g proven)

System B — Maglev + Laser-Thermal Cargo (build second)

- **Vehicle:** 5 t gross; maglev exit 2.0 km/s (6.8 km evacuated track, 30 g); laser-thermal stage: 6.8 km/s at Isp 900 on 2,685 kg LH₂; **payload 1,065 kg (21% — vs 2% chemical for the same job)**
- **Ground power:** 475 MW beam / 950 MW wall-plug for 462 s. Energy 122 MWh/launch (~\$12,200; **\$11/kg marginal**). 25 MW grid feed → 5 h recharge. Battery buffer note: 7.8C draw exceeds LFP comfort — oversize to ~500 MWh, use LTO, or hybridize with flywheels
- **Laser array:** ~24,000 × 20 kW fiber modules; \$2.4–9.5B at \$5–20/W (the dominant capex; riding a falling cost curve)
- **Beam path:** direct slant for early burn only; **relay mirrors are load-bearing** for the back half — 4–8 m passive mirrors, 100–500 kg each, 2–3 for full-burn coverage. Ground beam exits vertically (blooming N_D 83 → 28 with 300 m array footprint; relay drift 83 μrad/s is trackable). Relay constraint: ≥99.99% dielectric coatings (bench-testable). Alternative: 2–3 extra full ground arrays

downrange (~1000× more expensive)

- **Capex: \$4-12B.** Per-kg: \$36,600 @ 10 flights/yr; \$3,800 @ 100 (Falcon 9 parity); **\$870 @ 500; \$323 @ 2,000**

System C — Crewed Variant (build last)

- 6 seats (~1 t/seat), 3 g limit → maglev only 1.5 km/s on a 38 km track; 40 t gross; **3.8 GW beam**
- Crew *status*, not crew count, drives cost (g-limits, human-rating margin, abort). 6-max saves little; larger cabins amortize better
- **Abort:** beam loss = engine loss → independent solid chemical abort motor. Chemical as parachute, not propulsion
- **Capex \$23-84B.** Per seat: \$45M @ 10 flights/yr (market parity) → \$2.4M @ 200 → \$0.6M @ 1,000. An airline: ruinous at low utilization, unbeatable at high

Node D — LEO Dock + Tugs

- Orbit is **92% of escape energy**; the tug adds only 3.2 km/s. Docking at ~0 m/s relative replaces hypersonic catching; slugs aggregate into assembled ships
- **Tug menu** (propellant itself gun-delivered at ~\$150/kg — the depot bootstrap):

Tug	Isp	Trip	Propellant kg per cargo kg
Chemical hydrolox	450	3 days	1.07
Beamed-thermal (perigee kicks from the existing array — Oberth optimal)	900	~1 week	0.44
Electric Hall (spiral penalty: 7.7 km/s dv, still frugal)	2,500	6-9 mo	0.37

- Staged escape: \$216-320/kg vs \$190 direct — direct gun wins for dumb bulk; staging wins for anything that must arrive as a functioning spacecraft

Technology Risk Register & Retirement Path

Risk	Why it gates	Retirement path	Cost to retire
Hydrogen heat exchanger	Wall temp caps gas temp caps Isp caps payload. 48 MW/m ² absorbed (3× fusion divertor); 2,900 K hydrogen; walls sub-mm (microchannels mandatory); hot-H ₂ corrosion (NERVA's killer — fixed with ZrC/NbC; modern stack: TaC/HfC on carbon-carbon, whose 44 MPa thermal stress beats W-Re 16×)	Coupon (10 kW laser, \$10k) → tile (100 kW) → panel (1 MW pilot). Flux, not power, is what material sees — bench results transfer to 475 MW. Watch: Ledinegg flow instability (orifice the channels), 300→3,000 K cycle fatigue, hot spots at 1.5-2× design flux from AO residual	\$10k- \$5M
Design margin	Isp 900 sits at the solid-HX ceiling; laser-sustained plasma (gas hotter than walls) offers Isp 1,100-1,400 as block upgrade	De-rate check: Isp 850 → payload 21%→19%. Architecture robust	—
Beam combining + propagation	24k fibers into one phased aperture; thermal blooming self-defocus	Split-step wave-optics simulation (weekend of code), then subscale field tests. Relay-vertical geometry buys ~10× margin	~\$0 → \$10M
Relay mirror coatings	475 MW × 0.01% absorption = 48 kW radiative-only rejection	≥99.99% dielectric coating coupons, ground vacuum test	\$1- 10M
Pulsed power (gun)	36-64 GW / 1.3 s; nothing built within 1,000×	Flywheel/pulsed-alternator farm; subscale gun (Navy-railgun-class energies) first	\$50- 500M
Slug terminal ops	Kick stage must survive 1,000 g (guided-artillery heritage); deep-space catch unpriced for direct-escape variant	Staged-to-LEO architecture sidesteps it (tug docks, doesn't catch)	design choice

Demonstration Ladder

Phase	Beam / hardware	Delivers	Cost	Milestone
0	Benchtop	—	<\$1k	Inductrack drum + Halbach cart (below); watt-class lightcraft
1	10 kW laser	100 g craft	~\$100k	Lightcraft-class climb (Myrabo flew this in 2000: 50 g to 71 m)
1.5	100 kW (owned OR rented — industrial fiber / HELSTF beam time via SBIR customer)	1 kg free-flight hop	\$0.5- 3M	Suborbital-class demo at SBIR scale; closes the 100 g → 10 kg gap
2	1 MW array	10 kg suborbital	\$10- 50M	HX qualification at full flux ; AO field data. Buy the array only when flight cadence justifies capex — beam time rents before it owns
3	10 MW array	22 kg near-orbital	\$50- 200M	Venture-scale pilot; first revenue (microsats)
4	475 MW + 6.8 km track	1,065 kg LEO	\$4-12B	System B operational; coilgun (System A) built in parallel

5	3.8 GW + 38 km track	6 crew	\$23–84B	System C, on System B's flight heritage
---	----------------------	--------	----------	---

Benchtop Inductrack build (Magnetic Matter Movers catalog piece / scale model of the launch sled): $8 \times \frac{1}{2}$ " N42 cubes in Halbach ($\lambda = 51$ mm, $B_0 \approx 0.93$ T), 3D-printed carrier (N-red/S-blue pole language), 30 cm aluminum drum ≥ 3 mm wall, 50 W motor. Lift-off at ~ 54 RPM; ~ 1 cm hover gap at 1,000–1,500 RPM; constant ~ 28 W drag at any speed (the launch-relevant signature). Gotchas: Halbach assembly needs a clamping jig (adjacent magnets fight the pattern); add guide rails or V-groove drum for lateral stability; cap at 1,500 RPM for rim safety. BOM $\sim \$80$ –120.

Key Reference Numbers

- Delta- v to LEO: 9.4 km/s (7.8 orbital + ~ 1.6 losses) · Escape from surface: 11.2 · Escape from LEO: +3.2
- Isp 900 laser-thermal hydrogen: $T_{\text{gas}} \approx 2,900$ K; propellant fraction for 6.8 km/s = 54% (vs 86% chemical)
- Beam requirement: ~ 95 MW per tonne of vehicle gross (T/W 1.5, 50% coupling)
- Halbach near-field: 1 T at 1 cm gap ≈ 30 t/m² lift — $10^8 \times$ Earth-field pressure
- Inductrack L/D $\approx v/w$; drag *power* constant with speed (magnetic lift gets cheaper as you go faster; aero drag grows as v^3)
- Earth's field: 1 mPa pressure, 1 mJ/m³ — an orbital reboost resource only (ED tether: 1,555 m/s per month on 5 t)

Open Work Items

1. Split-step wave-optics propagation sim (turbulence screens + blooming feedback) — the go/no-go on beam delivery
2. Benchtop builds: Inductrack drum (build sheet pending); watt-class lightcraft demo
3. HX coupon test spec: channel geometry, instrumentation, hydrogen loop, target flux profile
4. Market/utilization model: flight-rate demand curves that set the break-even dates
5. Relay constellation geometry: orbits, count, and handoff timeline for 462 s coverage
6. Pulsed-power farm point design: flywheel count, discharge sequencing, subscale demonstrator spec

Addendum — Node D Revision: The Water Depot (Jul 15, 2026)

Tug trade result (full accounting: water at \$150/kg LEO, plant amortization, fleet sizing for 2,287 t/yr):

Tug	Water kg / cargo kg	Fleet needed	\$/kg to escape
Beamed steam (Isp 400, perigee kicks) — SELECTED	1.38	8	\$225
Hydrolox chemical (Isp 450)	1.36 (after 0.78 electrolysis yield)	4	\$269
Electric water plasma (Isp 800, spiral)	1.67	133	\$560

Key finding: steam and hydrolox consume essentially identical water per kg of cargo — electrolysis yield cancels the Isp advantage. The 192 t propellant plant (incl. 89 t of cryocoolers, the node's hardest development) buys nothing and is **deleted from baseline**. Node D reduces to: water storage tanks + docking ports + ~ 8 beamed-steam tugs. Hydrolox remains a fast-lane option (3-day vs 10-day trips) if a time-critical market appears; electric tugs lose on fleet capital (210-day trips), niche only.

Engineering delta: steam oxidizes carbon — tug HX uses oxide-coated or Ir-lined refractory instead of the C-C/TaC hydrogen stack. Same coupon test rig; add steam coupons to the campaign.

Architecture-wide principle, confirmed at every node: ship the dumbest possible mass; add energy at the point of use. Gun throws water; beam superheats it; one laser array serves ascent and perigee departure; one HX technology family (two coating systems) powers both.

Open work items — revised: - $\sim \sim$ Depot plant point design $\sim \sim \rightarrow$ deleted with the plant - Steam-HX coupon set added to item 8 (coatings vs. hot H₂O at 3,000 K) - New: steam tug point design — water tankage, perigee-kick pass count, beam scheduling against launch operations

Extension — The Mars Pipeline (Jul 15, 2026)

Principle: interplanetary space cannot hold a track — everything placed there is itself in solar orbit. The road (Kepler coasting) is free; all cost lives at the ramps. Infrastructure therefore replaces per-trip propellant only at the endpoints.

On-ramp: already built

The escape gun's 12.53 km/s atmospheric-exit velocity gives $v_\infty = 5.65$ km/s — Mars Hohmann needs only 2.95 (ground requirement 11.57 km/s). Margin buys 4–5 month fast transits and stretches each launch window to ~ 4 –6 months. Departure propellant: zero.

Coast: 259 days (Hohmann) / ~ 130 –150 (fast), propellant zero

Windows every 780 days (synodic). At 5,000 shots/yr, one window ships ~1,300 t Marsward; between windows the gun serves LEO/lunar traffic.

Off-ramp: aerocapture (baseline)

- Entry speeds 5.7 km/s (Hohmann) to 7.1 km/s (fast) — below MSL’s 5.8 for freight, well below Earth-return capsule heritage (12.5)
- Slug carries 12–15% ablative heatshield (1,000 g-compatible carbon nose, same family as launch ablator)
- Aerocapture pass → Mars orbit → 50–100 m/s periapsis-raise (steam puff)
- Arrival propellant: ~zero. Toll per kg ≈ heatshield mass fraction

Mars-orbit port (the ownable asset)

Tug fleet’s one-time emplacement job, then Earth tugs retire from the route: station + water stock, nav/comm beacons, small local steam tugs (watered by incoming slugs) for orbit consolidation. Nobody owns the transfer orbit; the port sets the price of mass at Mars.

Phase-2 upgrade: Phobos momentum bank

Tether/rotating catcher anchored to Phobos (1.07×10¹⁶ kg): inbound freight deposits momentum into Phobos’s orbit; outbound Earthward cargo withdraws it — inbound traffic funds outbound flings. 2,000 t/yr of 3 km/s catches perturbs Phobos by ~micrometers/s per year (planetary flywheel, effectively infinite bank). Passes the step-function filter as an *optional* upgrade: aerocapture runs the route without it. Softer than the Earth tether (no debris belt, lower closing speeds, free anchor mass); still inherits the catch-precision problem — phase it after the route is proven.

Steady state

Gun throws water and freight → Kepler carries → Mars atmosphere brakes → port aggregates. Per-trip consumables: one carbon nose and a puff of steam. The tug’s role, as specified: build the off-ramp once, then stand down.

New open work items: 13. Aerocapture corridor analysis: entry-angle tolerance vs. gun dispersion + mid-course trim budget (kick-stage sizing) 18. Mars port point design: station mass, local tug count, water buffer, emplacement flight manifest 19. Phobos catcher feasibility memo: catch tolerance vs. Mars-approach nav accuracy (phase-2 gate)

Extension — Destination Ranking (Jul 15, 2026)

Law: the gun and the Kepler coast are destination-agnostic; only the off-ramp changes. Destinations rank by their brakes: atmospheres are free off-ramps; airless bodies charge a steam toll — but airless + low-g bodies are where the coilgun re-roots and turns importers into exporters.

Destination	Ground req (km/s)	Gun margin	Transit	Window	Off-ramp	Toll per kg cargo
Moon	10.95	huge	3 d	daily	propulsive (steam)	0.24 kg water (self-paying when cargo is water)
Venus	11.46	+1.07	146 d	584 d	thick atmo — generous aerocapture corridor (entry 10.7 km/s)	~0.14 heatshield
Mars	11.57	+0.96	259 d	780 d	thin atmo — narrow corridor (work item 13)	~0.13 heatshield
Jupiter+	14.23	−1.70 (out of direct reach)	997 d+	399 d	—	via gravity assists or tug Oberth perigee kick

Sequence

1. **Moon first — the debug destination.** Daily windows delete the gun’s idle-capacity problem; 3-day feedback loop iterates operations ~250× faster than Mars; nearest-term freight market. Phase-2: lunar surface coilgun (escape 2.38 km/s → **290 m track at 1,000 g**; vacuum, low-g, no weather = ideal site) plus lunavator drive the off-ramp toll to zero and flip the Moon to exporter (polar water, regolith, metals). Earth launcher tech exports at 1/30 scale.
2. **Mars — the anchor tenant.** Full pipeline per prior extension; Phobos momentum bank as phase-2.
3. **Venus — the switch-yard.** Best transport physics (frequent windows, short transit, forgiving aerocapture), thinnest market (no reachable surface, no moon for momentum banking). Small port justified as gravity-assist junction to Mercury and outer-system routing, not as a terminus.
4. **Beyond the belt:** the gun’s direct horizon ends near Jupiter’s requirements; outer-system freight stages through tug perigee kicks (Oberth) or planetary assists.

New open work items: 14. Venus corridor memo: aerocapture tolerance vs. Mars (quantify the thin/thick corridor gap for work item 13 dispersion budget) 15. Lunar off-ramp point design: steam-brake tug at LLO vs. direct semi-hard bulk delivery; polar water bootstrap timeline 17. Lunar surface coilgun concept: 290 m track, power source (solar + flywheel), catch-mass economy with Earth pipeline

Extension — Demand Generation & the Slug Interface Standard (Jul 16, 2026)

Premise: every cost model in this study clears its break-even only at high flight rates; demand is therefore the binding constraint, ahead of any single technology. History shows demand in space has been created by *documents*, not consortia: a free interface standard (CubeSat spec, 1999 → the smallsat industry), a published price (rideshare price sheets → manifested smallsat demand), and an anchor-tenant purchase promise (CLPS → seven lander companies). Coordination through paper scales; coordination through meetings doesn't. The pipeline adopts the same instrument set.

Slug Interface Standard (SIS) — Draft A scope

A free, published specification letting any mission designer, CubeSat program, servicing startup, or depot operator design against the pipeline before it exists:

- **Mass classes & envelopes:** S-630 (baseline LEO slug, 630 kg net), S-670 (direct-escape, 670 kg), sub-classes for rideshare partitions within one slug
- **Environment rating:** 1,000 g axial design / 1,500 g proof; vibration & thermal transit envelopes (launch heat pulse, coast cold-soak)
- **Berthing interface:** standard grapple/berthing feature for tug capture at ~0 m/s relative; alignment, capture-envelope, and load-path definitions
- **Water payload spec:** purity classes (propellant-grade vs. life-support-grade), freeze state, tank venting standard
- **Kick-stage envelope:** volume/mass allocation, minimum reliability floor and disposal requirement (debris control at 1,000+ shots/yr is a licensing precondition)
- **Nav & transponder minimums:** tracking, arrival-corridor trim authority, abort-to-disposal logic
- **Heatshield classes:** L (none — LEO), M (Mars aerocapture, ~13%), V (Venus, ~14%)

Companion instruments

- **Published price sheet:** the \$/kg curves from this study, stated as offered service prices at declared flight rates — designers write proposals against numbers, not visions
- **Demand registry:** non-binding letters of intent (“would purchase N kg water at L1/LLO/Mars orbit at \$X/kg”) — the traction instrument that converts latent demand into evidence for funders
- **Circulation venues:** CONFERS (on-orbit servicing/assembly standards body), SmallSat conference, standards submission once Draft B stabilizes

Phase placement

SIS Draft A is a Phase 1–2 deliverable (cost ≈ \$0; authored alongside the wave-optics work). The registry opens with Draft A publication. Draft B incorporates feedback before any Phase 3 hardware commitment, so the 10 MW pilot flies payloads already designed to the standard.

New open work items: 5. Author SIS Draft A (spec skeleton above → full document with interface drawings) 6. Demand registry instrument: LOI template, pricing tiers, publication mechanism 7. Standards-body engagement plan: CONFERS membership path, SmallSat paper submission

Extension — External Validation & Design Imports (Jul 17, 2026)

Source: the Japanese 28 GHz microwave beamed-propulsion program (Shimamura et al. — J. Spacecraft & Rockets drone power-beaming test; Sci. Rep. 2025 tractor-beam and vortex-plate studies; Phys. Plasmas rectenna monitoring). The nearest running hardware program to this architecture's Phase 1; four items adopted:

1. **Tracking-collapse validation (risk register: beam combining + propagation).** Their measured end-to-end efficiency fell from 60.1% (fixed receiver) to 0.43% (free-flying drone) — a ~140× collapse purely from target motion. First published hardware confirmation of the tracking argument that makes relay mirrors load-bearing in this architecture. Cite in the propagation white paper.
2. **Beam shaping as thermal-control actuator (risk register: heat exchanger, hot spots).** Their vortex-phase-plate result showed beam *shape* determines where energy deposits and whether force helps or opposes the vehicle. Adopted implication: our phased array actively flattens the flux profile across the HX panel — the hot-spot problem (AO residual 1.5–2× local flux) gains an active mitigation path in addition to passive HX margin. Adds a requirement to Phase 2–3 array control software; adds a flux-profile command interface to the SIS-adjacent vehicle spec.
3. **Transmitter-as-sensor (risk register: HX; Build Sheet 002).** Their rectenna standing-wave monitor watched the plasma front without cameras. Adopted: backscatter monitoring of the HX panel from the ground array (per-subaperture return power = panel health map); at Phase 0, a photodiode watching droplet scatter.
4. **Beam-riding misalignment data (Build Sheet 002 §7).** Their offset-vs-thrust studies are the gyrotron-scale version of the droplet limit-cycle; the deliberate-offset protocol run added to Build Sheet 002 reproduces the measurement at bench scale.

Their unresolved problem we structurally avoid: pulsed air-plasma thrusters self-shield (residual plasma absorbs subsequent

pulses), forcing front-fed “tractor” geometries. The CW solid-HX vehicle has no plasma and no pulse timing — a design rationale line for the white paper.

New open work item: 11. Flux-profile control requirement: define commanded beam-shape modes (flat-top, edge-weighted, hot-spot-notch) and the HX health feedback loop closing through per-subaperture backscatter.

Extension — Wave-Optics Propagation: First-Cut Results (Jul 19, 2026)

Work item 1 status: **first-cut split-step simulation complete** (script: `mmm_waveoptics_v0.py`, MIT-licensed). Method: Kolmogorov/von Karman phase screens (structure-function-calibrated) layered per the HV5/7 profile from the 2 km site; AO modeled physically as conjugation of the beacon-path column with 10 cm actuator-pitch fitting cutoff; **point-ahead anisoplanatism implemented by laterally shifting each turbulence layer by θ_{pa-h} for the outgoing beam**; vacuum far-field beyond the atmosphere; 8-draw Monte Carlo; metric = power fraction into the 2.5 m HX at range.

Case	Zenith	Range	Point-ahead	AO	Delivered fraction
Vertical → relay mirror (baseline)	0°	600 km	2 μrad	on	0.83 ± 0.01
Slant mid-burn → vehicle	55°	350 km	20 μrad	on	0.64 ± 0.05
Vertical, no AO	0°	600 km	—	off	0.46 ± 0.09
Slant, no AO	55°	350 km	—	off	0.56 ± 0.09

(Diffraction-limited/no-atmosphere reference ≈ 0.99. Slant bucket angles are geometrically larger at closer range; comparison is against each case’s own delivery requirement.)

Findings. (1) The **relay-vertical baseline closes**: 83% delivery through the full atmosphere with modest AO, tight variance — before array-scale refinements, which only help. (2) **AO is structurally ineffective on the slant path** (0.56 → 0.64): with the correction measured through air offset from the beam’s path by the point-ahead angle, conjugation decorrelates — now confirmed by wave propagation, independently of the earlier isoplanatic-patch scaling argument and the published 60.1%→0.43% moving-target collapse. Shown slant numbers *flatter* reality: end-of-burn (72°, 700 km) is worse on every term and carries blooming $N_D = 270$ (excluded from this model; vertical $N_D = 28$ with 300 m footprint).

Vertical phased-array configuration confirmed as baseline. The simulated 1 m director is a single subaperture of the distributed array; array synthesis (many subapertures over a 100-300 m footprint) is deferred to the full Phase 1 sim — with the standing correction that **thermal absorption requires no optical coherence**: incoherent tiled addition suffices, so each subaperture needs only its own AO and pointing, and array-wide phase-locking is deleted from the requirements entirely, but distributes per-aperture flux and *reduces* blooming — the first-cut result is therefore conservative for the baseline and optimistic only for the already-rejected slant case.

Gauntlet update (v1, same date): blooming coupled into propagation (similarity-scaled steady-state upwind-integral model, iterated to convergence) plus AO temporal lag (1 ms latency against Bufton frozen-flow winds). Results: **vertical relay baseline 0.75 ± 0.02 with the full degradation budget** (blooming-off reference 0.80); slant progression 0.78 (early, 30°) → 0.22 (mid, 55°) → **0.03 (end-of-burn, 72°, N_D 270)** — quantitatively reproducing the “direct works early, relays carry the back half” design claim from an independent method. Script: `mmm_waveoptics_v1.py`. Residual caveat on the baseline: phase-compensation instability not modeled (threatens high- N_D slant far more than N_D 28 vertical; explicit check assigned to Phase 1 completion).

Remaining Phase 1 sim scope (work item 1, continued): time-dependent thermal blooming coupled into propagation (the one term that could still move the vertical verdict); end-of-burn geometry sweep; Greenwood/temporal AO bandwidth error; distributed-array aperture synthesis; ≥100-draw Monte Carlo; radiometric absorption ledger.

Extension — Test Infrastructure & Range Plan (Jul 19, 2026)

Height-ladder finding: test altitude requirements are bimodal — bench height, then ~30 ft, then open sky. Intermediate tall-ceiling facilities (100-400 ft class) buy nothing: beam-riding control qualifies on **guide-wire tethered hops at 20-30 ft** (Myrabo lightcraft method), and free flight wants kilometers, not rafters. No 400 ft facility exists to rent, and none is needed.

The ladder (all within ~3 h of Rio Rancho — New Mexico is the one state where the full test infrastructure is a day trip):

Tier	Venue	Phase	What runs there	Access route
Bench	Home shop	0-1	Builds 001/002, torsion stand, low-power droplet	owned
Laser lab	Light-industrial rental, ABQ/Rio Rancho (~1-2k sq ft, 3-phase 240 V, interlocked blackout laser room, ventilation + exterior gas storage for hydrogen coupons; 14 ft ceiling suffices)	2	1-10 kW HX coupon campaign, steam microthruster, coilgun stage bench	~\$1-2k/mo lease at Phase 2 funding
Tethered hop	Same lab or 26-32 ft warehouse	2-3	Beam-riding control on guide wire	lease

Gun range	EMRTC (NM Tech, Socorro)	2-3	Subscale coilgun projectiles beyond lab backstop	commercial range rental
Free flight	Spaceport America	3+	10 kg vehicle flights; restricted-airspace umbrella eases outdoor-laser clearance	negotiated campaign contract
National labs	Sandia via NMSBA	1+	Pulsed power, magnetics, HX consultation at no cost to NM small businesses; CRADA path for facilities	apply after LLC formation
MW-class laser	White Sands / HELSTF	3+	Megawatt beam heritage (lightcraft flew here)	SBIR-customer partnership, not rental

Regulatory (file mentally now, act at Phase 2/3): outdoor beams require FAA coordination (AC 70-1) and DoD **Laser Clearinghouse** declassification for sky-directed propagation — the decisive argument for testing under Spaceport America’s restricted airspace rather than private land.

New open work items: 3. NMSBA application (immediately after LLC formation — free, slow, high-value) 4. Phase 2 laser-lab requirements sheet (power, interlock, ventilation, hydrogen storage) for the eventual lease search; Spaceport America first contact at Phase 3 planning 10. Phase 1 sim completion per remaining scope above (blooming coupling is the gate item)

Work Item Status Ledger (maintained live; items close when completed, without prompting)

Conventions: the ledger is ordered easiest → hardest, and IDs match that order 1-19. IDs were renumbered once to complexity order by owner decision, Jul 19, 2026, with all cross-references updated in the same edit; they are frozen henceforth. CLOSED items are immutable records. Last updated Jul 20, 2026.

Tier 1 — Desk & paper, current phase (0-1): items 1-10, hours to weeks each, ~\$0

#	Item	Status
1	Split-step wave-optics propagation sim	CLOSED — v0 + v1 gauntlet complete (0.75 vertical baseline; slant 0.78/0.22/0.03); remainder transferred to 10
2	Benchtop builds (Inductrack drum; droplet rig)	OPEN — build sheets released; hardware ordering underway
3	NMSBA application	OPEN — triggers immediately on LLC formation
4	Phase 2 laser-lab requirements sheet; Spaceport America first contact at Phase 3 planning	OPEN
5	SIS Draft A authoring	OPEN — skeleton published; full document is a Phase 1-2 deliverable
6	Demand registry instrument (LOI template, tiers)	OPEN — pairs with 5
7	Standards-body engagement plan (CONFERS, SmallSat)	OPEN — follows 5
8	HX coupon test spec (hydrogen + steam coupon sets)	OPEN — Phase 1; gates Phase 2 hardware; spec extended to ≥1,000-cycle automated endurance
9	Market/utilization model (flight-rate demand curves)	OPEN
10	Phase 1 sim completion (inherits from item 1): PCI stability check, distributed-array synthesis, ≥100-draw Monte Carlo, radiometric ledger	OPEN — blooming coupling and end-of-burn sweep already delivered by v1; scope reduced
20	Cadence endurance rig spec: automated subscale coilgun, 5,000-cycle campaign, condition monitoring + cost-per-shot ledger (~\$10-30k, Phase 2)	OPEN — new under cadence doctrine
21	Beamed-line availability analysis: NM cloud/wind statistics × Laser Clearinghouse windows → flyable-days model feeding the cadence economics	OPEN — first cut v0 published Jul 20 (mmm_item21_flyable_days_v0, this directory): ~200 flyable days/yr (150-250); LCH quantified from published observatory data; availability binds at the 2,000-flights/yr tier; v1 refinement scope listed therein
22	Citation & uncertainty pass: source every cost figure, band every result, apply [E]/[I]/[S] labels line-by-line per the Epistemic Convention — precondition for journal submission AND now mandated by owner’s accuracy standard	OPEN — kickoff published Jul 20 (mmm_item22_citation_pass_v0, this directory): 11 headline rows sourced; corrections applied to this document in the same commit; full inference-chain pass remains
23	Tube exit-aperture concept: fast shutter vs. plasma window vs. per-shot membrane; km-scale thermal expansion joints	OPEN — red-team finding; Tier 2
24	Beam-loss range-safety footprint analysis + relay pointing-stability budget under thermal load	OPEN — red-team finding; Tier 2
25	Trademark application: “Magnetic Matter Movers” word mark via USPTO TEAS — base fee \$350/class (+\$200/class surcharge if using custom goods/services description instead of the ID Manual); file after LLC formation so the LLC is owner; likely classes: research/publication services + catalog hardware (one class suffices to start). Not deadline-bound: common-law rights accrue from use, and the dated public repo establishes first use	OPEN — new Jul 20, 2026

Tier 2 — Engineering analyses, Phases 1-3: items 11-16, weeks to months each, still simulation/paper

#	Item	Status
11	Flux-profile control requirement (beam-shape modes + backscatter loop)	OPEN
12	Relay constellation geometry (orbits, count, 462 s handoff)	OPEN
13	Aerocapture corridor vs. gun dispersion + kick-stage trim budget	OPEN — first data-free Monte Carlo candidate
14	Venus corridor memo (quantifies thin/thick gap for 13)	OPEN — pairs with 13
15	Lunar off-ramp point design (steam brake vs. semi-hard bulk; polar water bootstrap)	OPEN
16	Pulsed-power farm point design + subscale demonstrator spec	OPEN — hardest Tier 2 item; feeds System A; now carries 5,000-cycle switch/coil/flywheel life requirement

Tier 3 — Far-horizon concept work, Phases 3+: items 17-19, meaningful only after earlier tiers return data

#	Item	Status
17	Lunar surface coilgun concept (290 m track; catch-mass economy)	OPEN
18	Mars port point design (station mass, tug count, emplacement manifest)	OPEN
19	Phobos catcher feasibility memo	OPEN — deliberately last: gated behind a proven Mars route

Extension — Environmental Ledger & Cadence Doctrine (Jul 19, 2026)

Environmental ledger (thesis-relevant: constraint as moat)

Operational emissions are electricity: 115 MWh/t (beamed) and 21 MWh/t (gun) — near-zero carbon from NM solar; exhaust is steam; no soot, alumina, perchlorates, hypergolics, or deluge contamination anywhere in the stack. Chemical launch, by contrast, carries ~25-30 t CO₂/t payload plus ~0.3-3 t black carbon per launch (engine/cycle-dependent) injected at stratospheric altitude (≈500× surface warming potency — Ryan et al. 2022, sourced) — a compounding regulatory exposure that grows with exactly the industry-wide cadence this program bets on. Tightening launch regulation is a tailwind for this architecture.

Honest residuals, ranked: (1) sonic booms — the dominant real externality at 20+ shots/day; priced into WSMR-adjacent siting; (2) ~24 t H₂O per beamed launch partly above the tropopause — monitor-and-report item, owned in the white paper preemptively; (3) ~1 kg ablated carbon per gun shot (≈5 t/yr at full cadence — negligible vs. aviation); (4) beam-path wildlife/aviation safety — radar-triggered beam interrupt, Phase 3 requirement.

Cadence doctrine (gate plan amended same date)

Economics live at 100+ flights/yr and 1,000-5,000 shots/yr; gates must therefore prove **cycling, not just capability**. Cadence risk is cycle-count-invariant — switch lifetime, coil thermal fatigue, flywheel cycling, HX cycle life all retire at benchtop scale. Adopted: Phase 2 cadence endurance rig (item 20: 5,000 automated subscale cycles, ~\$10-30k); HX coupon protocol extended to ≥1,000 cycles; Phase 3 gate adds sustained flights/week; Phase 4 adds dummy-payload cadence campaigns at marginal (energy-only) cost. **Low-demand operating mode is designed, not feared:** 50 bookings + 450 water-slug dummy shots ≈ a 500-shot year for ~\$6k of additional electricity — the cycle ledger accumulated on unsold capacity is itself the sales instrument. No chemical system can afford to prove cadence on empty manifests; this one barely notices the cost.

Extension — Red-Team Register (Jul 19, 2026)

Adversarial self-review ahead of any public submission. **Hidden advantages banked:** (1) thermal propulsion needs incoherent tiled power, not a phased beam — array-wide coherence deleted from requirements (correction applied above); (2) unburned slugs are suborbital by default — orbit requires an affirmative kick, so the gun's failure mode is disposal, not debris; (3) the 500 MWh battery farm is a merchant grid-storage asset between launches; (4) gun (all-weather) and beam (clear-sky) hedge each other's availability; (5) engineless vehicles make marginal airframe cost trivial.

Hidden liabilities opened as work items: beamed-line weather/clearance availability unquantified (21); tube exit aperture and km-scale expansion unaddressed (23); relay μrad pointing under load likely 2-3× mass growth, and cargo beam-loss impact footprints undrafted (24); dual-use/ITAR posture for a steerable near-GW array to be decided deliberately, before external framing decides it for us.

Peer-review posture: the study survives review as a *concept/architecture paper* (Acta Astronautica / JBIS / AIAA class) contingent on item 22 — citations on every cost figure, uncertainty bands on every result, claims scoped to evidence. It is not, and does not claim to be, a validated engineering study; the gate/kill-criteria apparatus is the submission's differentiating strength and should be foregrounded.

Extension — Cost Calibration Doctrine (Jul 20, 2026)

Motivation. The item-22 citation pass showed that memory-sourced cost errors are not uniformly directional and therefore cannot be fixed by a single “% of realism” dial: the chemical-launch comparison was stale in the *unfavorable* direction (published 2026 prices are lower than the strawman), the EM-launcher extrapolation ratio was ~5% high, and the black-carbon figure was fine at midpoint but banded too narrowly. Meanwhile the published record on first-of-a-kind megaprojects (pioneer-plant and aerospace overrun literature) puts early capital estimates at roughly **40-60% of final actuals** — 2-3× growth is the historical norm, not the exception. A program that publishes optimistic points is betting reviewers won’t know this; they will.

The doctrine (amends the Epistemic Convention; owner-adopted):

- 1. **Physics exact.** No calibration dial ever applies to [E]-tier claims.
- 2. **Costs are bands, not points.** The stated midpoint is the honest estimate — never a target, never a goal-seeded number. Default band ×2-3 for first-of-a-kind items until item 22 closes that figure with sources; quotes and bids narrow bands, opinions don’t.
- 3. **The 3× survivability test.** Every headline \$/kg or capex claim carries the question: *does the architecture keep its advantage over the incumbent at 3× the estimate?* Advantage retained at 3× = the claim is load-bearing; advantage inverted inside the band = the claim is decorative and must not carry an argument.
- 4. **Kill-criterion restatement.** A cost estimate “fails” when the *advantage inverts inside the band* — not when the midpoint moves. An estimate that doubles while keeping a 10× advantage is a pass and is reported as such.

The test applied to current headline claims:

Claim	Midpoint	At 3×	Advantage at 3×
Gun freight \$/kg to LEO	\$150	\$450	Yes — vs \$3-7k/kg published chemical; ~10× margin survives
Beamed cargo \$/kg (500-2,000 flights/yr)	\$323-870	~\$1,000-2,600	Yes at rate — still at/below Falcon parity; but break-even flight rate ≈3× higher → item 9 (demand model) is the binding analysis, not the capex digit
Marginal energy cost	\$11/kg	\$33/kg	Trivially — remains noise against any incumbent
Phase 0-2 program cost	\$270-\$500k	scales	Absorbable at self/SBIR funding tiers by design
Crew capex (System C)	\$23-84B	\$69-252B	No — not robust at 3×; consistent with the already-accepted end state (gate plan Phase 6 kill: chemical crew taxis remain the permanent answer)

Reading: the architecture survives the realistic number everywhere the program actually bets. The one claim that fails the 3× test (crew) already carries its own kill criterion and an acceptable end state — which is the correct shape for a failing cost claim: pre-declared, gated, and non-load-bearing. This table, maintained as estimates are re-sourced under item 22, is itself the anti-fragility exhibit for reviewers and funders: the program does not need its optimistic numbers to be right.